

Given an acute triangle ABC with $\angle CBA = 75^\circ$, orthocenter H , and circumcircle Ω , we need to determine the largest real number x such that for sufficiently large i , $n_{2i} \geq y \cdot x^i$ and the corresponding z for the same inequality. The final expression $x + 2016z$ must be written in the form $p + q\sqrt{r}$, and we need to find $p + q + r$.

Key steps and reasoning:

1. **Problem Analysis:** The problem involves complex geometric configurations and number theory. The sequence of amazing numbers n_i grows exponentially. The ratio $\frac{CP}{CH} = \sqrt{n}$ and integer lengths AQ and H_CQ suggest trigonometric and algebraic relationships.
2. **Trigonometric and Algebraic Considerations:** The angle 75° involves trigonometric identities with $\sqrt{3}$ and $\sqrt{6}$. The growth rate of the sequence n_i is tied to the fundamental unit in the ring $\mathbb{Z}[\sqrt{3}]$, which is $2 + \sqrt{3}$. The norm of this unit squared, $(2 + \sqrt{3})^2 = 7 + 4\sqrt{3}$, suggests the growth rate $x = 7 + 4\sqrt{3}$.
3. **Determining z :** The parameter z is the largest real number such that $n_{2i} \geq z \cdot x^i$. Given the exponential growth, z is taken as 1, the minimal constant factor.
4. **Final Calculation:** The expression $x + 2016z$ becomes $7 + 4\sqrt{3} + 2016 \cdot 1 = 2023 + 4\sqrt{3}$. Here, $p = 2023$, $q = 4$, and $r = 3$. Summing these values gives $2023 + 4 + 3 = 2030$.

2030